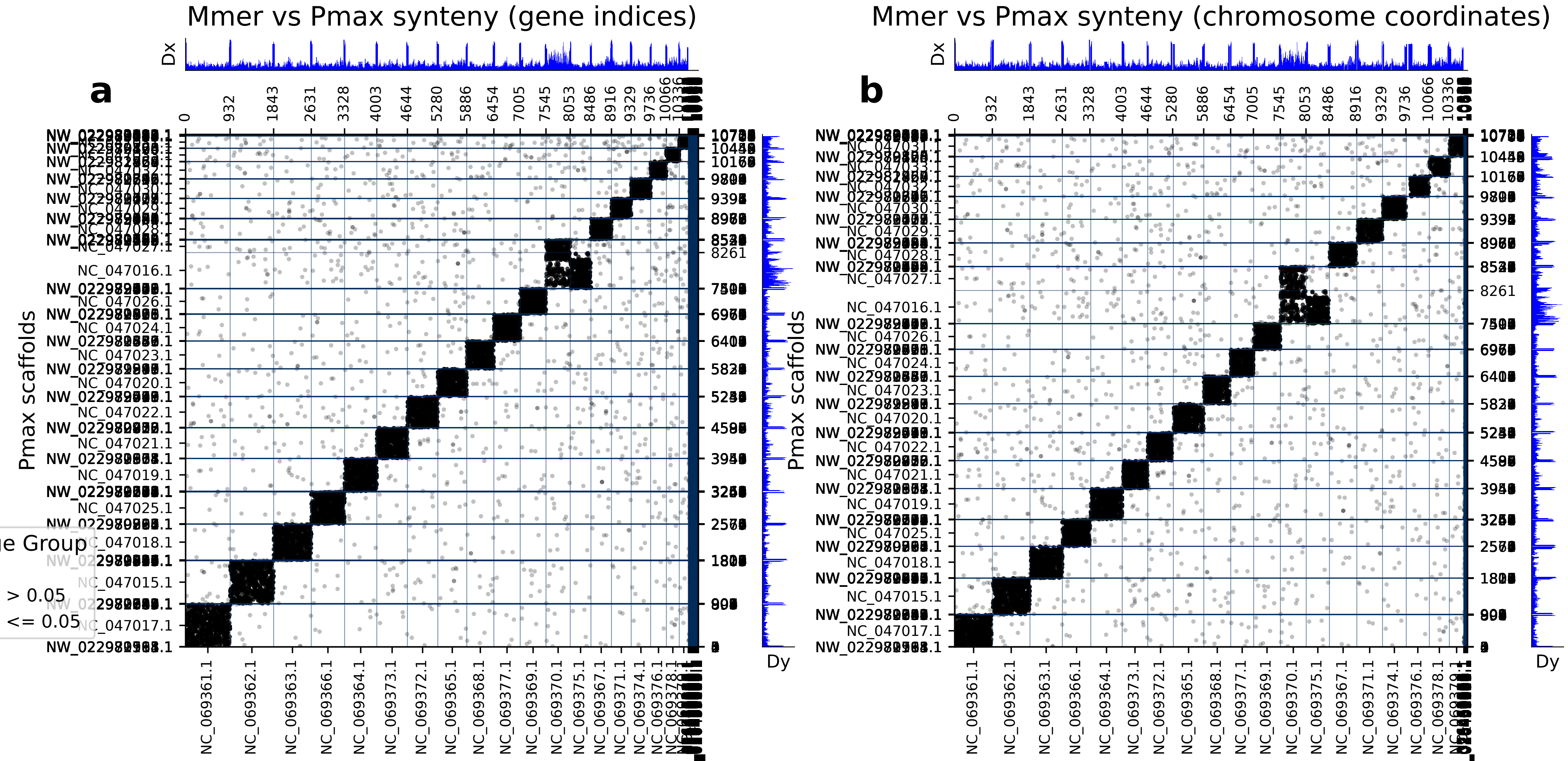
diamond: 10737 orthologs on 214 Mmer scaffolds and 237 Pmax scaffolds.



c

Mmer vs Pmax groups

Mmer scaffolds

d

Mmer vs Pmax FET, whole chroms.

	Mmer scaf	Pmax scaf	p (FET)	Count
0	NC_069361.1	NC_047017.1	0.00e+00	801
1	NC_069362.1	NC_047015.1	0.00e+00	788
2	NC_069363.1	NC_047018.1	0.00e+00	689
3	NC_069366.1	NC_047025.1	0.00e+00	585
4	NC_069364.1	NC_047019.1	0.00e+00	583
5	NC_069373.1	NC_047021.1	0.00e+00	542
6	NC_069372.1	NC_047022.1	0.00e+00	525
7	NC_069365.1	NC_047020.1	0.00e+00	496
8	NC_069368.1	NC_047023.1	0.00e+00	492
9	NC_069377.1	NC_047024.1	0.00e+00	477
10	NC_069369.1	NC_047026.1	0.00e+00	452
11	NC_069375.1	NC_047016.1	0.00e+00	368
12	NC_069367.1	NC_047028.1	0.00e+00	342
13	NC_069374.1	NC_047030.1	0.00e+00	334
14	NC_069371.1	NC_047029.1	0.00e+00	330
15	NC_069376.1	NC_047032.1	0.00e+00	257
16	NC_069370.1	NC_047016.1	3.11e-147	241
17	NC_069378.1	NC_047033.1	1.63e-283	196
18	NC_069370.1	NC_047027.1	1.74e-175	176
19	NC_069379.1	NC_047031.1	2.08e-193	137
20	NC_048487.1	NW_022980724.1	2.46e-07	3
21	NW_026462340.1	NW_022979575.1	7.92e-03	2

e

Mmer vs Pmax FET, chrom. pieces

Mmer scaffolds

	Mmer scaf	Pmax scaf	p (FET)	Count
0	NC_069361.1 : 0-end	NC_047017.1 : 0-end	0.00e+00	801
1	NC_069362.1 : 0-end	NC_047015.1 : 0-end	0.00e+00	788
2	NC_069363.1 : 0-end	NC_047018.1 : 0-end	0.00e+00	689
3	NC_069366.1 : 0-end	NC_047025.1 : 0-end	0.00e+00	585
4	NC_069364.1 : 0-end	NC_047019.1 : 0-end	0.00e+00	583
5	NC_069373.1 : 0-end	NC_047021.1 : 0-end	0.00e+00	542
6	NC_069372.1 : 0-end	NC_047022.1 : 0-end	0.00e+00	525
7	NC_069365.1 : 0-end	NC_047020.1 : 0-end	0.00e+00	496
8	NC_069368.1 : 0-end	NC_047023.1 : 0-end	0.00e+00	492
9	NC_069377.1 : 0-end	NC_047024.1 : 0-end	0.00e+00	477
10	NC_069369.1 : 0-end	NC_047026.1 : 0-end	0.00e+00	452
11	NC_069375.1 : 0-end	NC_047016.1 : 0-end	0.00e+00	368
12	NC_069367.1 : 0-end	NC_047028.1 : 0-end	0.00e+00	342
13	NC_069374.1 : 0-end	NC_047030.1 : 0-end	0.00e+00	334
14	NC_069371.1 : 0-end	NC_047029.1 : 0-end	0.00e+00	330
15	NC_069376.1 : 0-end	NC_047032.1 : 0-end	0.00e+00	257
16	NC_069370.1 : 0-end	NC_047016.1 : 0-end	3.11e-147	241
17	NC_069378.1 : 0-end	NC_047033.1 : 0-end	1.63e-283	196
18	NC_069370.1 : 0-end	NC_047027.1 : 0-end	1.74e-175	176
19	NC_069379.1 : 0-end	NC_047031.1 : 0-end	2.08e-193	137
20	NC_048487.1 : 0-end	NW_022980724.1 : 0-end	2.46e-07	3
21	NW_026462340.1 : 0-end	NW_022979575.1 : 0-end	7.92e-03	2

(a) depicts the chromosomes/scaffolds of Mmer (x-axis) plotted against the chromosomes/scaffolds of Pmax (y_axis). Each dot in the plot represents an ortholog, specifically a reciprocal best diamond blastp match between two species. The unit of the x- and y-axes are the number of orthologous proteins between these two species: 10737 orthologs found between 214 Mmer scaffolds and 237 Pmax scaffolds. If there are chromosome breaks, Fisher's exact test is used to calculate the significance of the interactions between the sub-chromosomal pieces. Otherwise Fisher's exact test is calculated on whole chromosomes. The opacity of the dots depict the significance from Fisher's exact test. Dots that are a solid color are in cells with a FET p-value less than or equal to 0.05. Dots that are translucent are in cells with a FET p-value greater than 0.05. The Dx and Dy values depict places where there may be sudden breaks in synteny. See the supplementary information the following paper for more information on Dx and Dy: Simakov, Oleg, et al. Deeply conserved synteny resolves early events in vertebrate evolution. Nature Ecology & Evolution 4.6 (2020): 820-830. (b) depicts the same information as panel a, but it is plotted in the organisms chromosome basepair coordinates rather than gene index. This is useful for visualizing gene-poor regions of the chromosomes (c) shows which gene groups are most prevalent in each chromosome pair. The FET p-value in this table corresponds to the whole-chormosome FET p-value , and is not a FET value of the correlation between the chromosome pair and the gene group. (d) shows chromosome-scale significance values. This table shows the same information as c, but does not factor in gene group information. (e) shows the FET p-values of the sub-chromosomal compartments. Useful for showing if single arms of chromosomes are correlated with other regions.